Task 2: Data Retrieval & Sorting Dashboard

Aim

To design and develop a dashboard that retrieves and displays student (or employee) records with functionalities for sorting, filtering, and counting data based on specific attributes such as name, date, and department.

Algorithm

1. **Start**
2. **Data Collection**
    - Retrieve student/employee records from a data source (database, API, or static dataset).
    - Each record should include:
        - Name
        - Date (e.g., joining date or DOB)
        - Department
3. **Display Data**
    - Render the records in a tabular format on the dashboard.
4. **Sorting Functionality**
    - Provide options to sort:
        - By Name (ascending/descending)
        - By Date (ascending/descending)
    - When a sort option is selected:
        - Apply sorting logic on the dataset.
        - Update the displayed table.
5. **Filtering by Department**
    - Create a dropdown or selection menu for departments.
    - When a department is selected:
        - Filter records matching the chosen department.
        - Refresh the table with filtered data.
6. **Count Calculation**

- Group records by department.
- Count the number of students/employees in each department.
- Display the counts (e.g., summary section or chart).

7. **Dynamic Updates**
   - Ensure sorting and filtering work together dynamically.
   - Update counts based on filtered data if required.
8. **End**

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Staff Directory</title>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-slate-50">
  <div class="min-h-screen p-4 md:p-8 font-sans text-slate-900 relative">
    <div class="max-w-6xl mx-auto">
      <header class="mb-8 flex flex-col md:flex-row md:items-center justify-
between gap-4 border-b border-slate-200 pb-6">
        <div>
          <div class="flex items-center gap-3 mb-1">
            <div class="bg-indigo-600 p-2 rounded-lg text-white text-2xl">👥</div>
            <h1 class="text-3xl font-bold text-slate-800 tracking-tight">Staff
Directory</html>
            </div>
          <div class="flex items-center gap-3 mt-2">
            <p class="text-slate-500 text-sm">Real-time Cloud Management</p>
```

```
        <div id="connectionStatus" class="px-2 py-0.5 rounded-full text-[10px] font-bold uppercase tracking-widest flex items-center gap-1.5 bg-amber-100 text-amber-700">
          <div class="status-dot w-1.5 h-1.5 rounded-full bg-amber-500"></div>
          <span class="text-xs">🖫</span>
          <span class="status-text">Connecting</span>
        </div>
      </div>
    </div>
    <div class="flex gap-2">
      <button id="exportBtn" class="flex items-center gap-2 bg-white border border-slate-200 hover:bg-slate-50 text-slate-700 px-5 py-2.5 rounded-xl transition-all shadow-sm active:scale-95 font-medium">
        <span class="text-lg">⭳</span><span>Export</span>
      </button>
      <button type="button" id="addEmployeeBtn" class="flex items-center gap-2 bg-indigo-600 hover:bg-indigo-700 text-white px-5 py-2.5 rounded-xl transition-all shadow-md hover:shadow-indigo-200 active:scale-95 font-medium">
        <span class="text-lg">+</span><span>Add Employee</span>
      </button>
    </div>
  </header>
  <div id="errorDiv" class="mb-6 p-4 bg-red-50 border border-red-200 text-red-700 rounded-xl flex items-center gap-3 hidden">
    <span class="text-xl">⚠</span>
    <span class="text-sm font-medium"></span>
  </div>
  <section class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-8">
    <div class="bg-white p-6 rounded-2xl border border-slate-200 shadow-sm transition-transform hover:scale-[1.01]">
      <div class="flex items-center justify-between mb-2">
```

```
        <span class="text-slate-500 text-xs font-bold uppercase tracking-wider">Total Staff</span>
        <span class="text-indigo-500 text-xl">👥</span>
      </div>
      <div id="totalStaffCount" class="text-3xl font-bold">0</div>
    </div>
    <div id="statsContainer"></div>
  </section>
  <div class="bg-white p-4 rounded-t-2xl border border-slate-200 flex flex-col md:flex-row gap-4 items-center justify-between shadow-sm">
    <div class="flex flex-col md:flex-row gap-4 w-full md:w-auto items-center">
      <div class="relative w-full md:w-80">
        <span class="absolute left-3 top-1/2 -translate-y-1/2 text-slate-400 text-lg">🔍</span>
        <input id="searchInput" type="text" placeholder="Search by name or role..." class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm" />
      </div>
      <div class="relative w-full md:w-64">
        <span class="absolute left-3 top-1/2 -translate-y-1/2 text-slate-400 text-lg">💾</span>
        <input id="fileNameInput" type="text" placeholder="File name for export..." value="staff_directory_report.json" class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm italic" />
      </div>
    </div>
    <div class="flex items-center gap-3 w-full md:w-auto">
      <div class="relative w-full md:w-56">
        <span class="absolute left-3 top-1/2 -translate-y-1/2 text-slate-400 text-lg">▣</span>
```

```
        <select id="deptSelect" class="w-full pl-10 pr-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl appearance-none focus:outline-none focus:ring-2 focus:ring-indigo-500 transition-all cursor-pointer text-sm font-medium"></select>

        <span class="absolute right-3 top-1/2 -translate-y-1/2 text-slate-400 pointer-events-none text-sm">▼</span>

      </div>

    </div>

  </div>

  <div id="tableContainer" class="bg-white rounded-b-2xl border border-t-0 border-slate-200 shadow-sm overflow-hidden">

    <div class="overflow-x-auto">

      <table class="w-full text-left border-collapse">

        <thead>

          <tr class="bg-slate-50/50 border-b border-slate-200">

            <th id="nameHeaderBtn" class="p-4 font-semibold text-slate-600 cursor-pointer group select-none text-sm"><div class="flex items-center gap-2">Name <span id="nameArrow" class="text-slate-300 group-hover:text-slate-400 text-sm">↕</span></div></th>

            <th class="p-4 font-semibold text-slate-600 text-sm">Department</th>

            <th class="p-4 font-semibold text-slate-600 text-sm">Role</th>

            <th id="joinedHeaderBtn" class="p-4 font-semibold text-slate-600 cursor-pointer group select-none text-sm"><div class="flex items-center gap-2">Join Date <span id="joinedArrow" class="text-slate-300 group-hover:text-slate-400 text-sm">↕</span></div></th>

          </tr>

        </thead>

        <tbody id="tableBody" class="divide-y divide-slate-100"></tbody>

      </table>

    </div>

  </div>
```

```
    <footer class="mt-6 flex flex-col md:flex-row justify-between items-center text-[11px] font-bold text-slate-400 px-4 gap-4">
      <p id="processingInfo" class="uppercase tracking-widest">Displaying 0 of 0 Secure Records</p>
      <div id="footerStats" class="flex flex-wrap justify-center gap-x-6 gap-y-2 uppercase tracking-widest"></div>
    </footer>
  </div>
  <div id="modal" class="hidden fixed inset-0 z-50 flex items-center justify-center p-4 bg-slate-900/60 backdrop-blur-sm">
    <div class="bg-white w-full max-w-md rounded-2xl shadow-2xl border border-slate-200 overflow-hidden">
      <div class="p-6 border-b border-slate-100 flex items-center justify-between bg-indigo-50/30">
        <div class="flex items-center gap-3 text-indigo-700">
          <span class="text-lg">➕</span>
          <h2 class="text-lg font-bold">Add New Employee</h2>
        </div>
        <button id="closeModalBtn" class="p-2 hover:bg-white rounded-full text-slate-400 hover:text-slate-600 transition-colors">✕</button>
      </div>
      <form id="modalForm" class="p-6 space-y-4">
        <div class="space-y-1.5">
          <label class="text-xs font-bold text-slate-500 uppercase tracking-wider">Full Name</label>
          <input id="nameInput" required type="text" placeholder="e.g. John Doe" class="w-full px-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm" />
        </div>
        <div class="grid grid-cols-2 gap-4">
          <div class="space-y-1.5">
```

```
          <label class="text-xs font-bold text-slate-500 uppercase tracking-wider">Department</label>

          <select id="deptInput" class="w-full px-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm font-medium cursor-pointer"></select>

        </div>

        <div class="space-y-1.5">

          <label class="text-xs font-bold text-slate-500 uppercase tracking-wider">Join Date</label>

          <input id="dateInput" type="date" class="w-full px-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm" />

        </div>

      </div>

      <div class="space-y-1.5">

        <label class="text-xs font-bold text-slate-500 uppercase tracking-wider">Job Role</label>

        <input id="roleInput" required type="text" placeholder="e.g. Senior Developer" class="w-full px-4 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:outline-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-sm" />

      </div>

      <div class="pt-4 flex gap-3">

        <button type="button" id="cancelBtn" class="flex-1 px-4 py-2.5 border border-slate-200 text-slate-600 font-bold text-sm rounded-xl hover:bg-slate-50 transition-colors">Cancel</button>

        <button type="submit" class="flex-1 px-4 py-2.5 bg-indigo-600 text-white font-bold text-sm rounded-xl hover:bg-indigo-700 shadow-md shadow-indigo-100 transition-all active:scale-95">Confirm Addition</button>

      </div>

    </form>

  </div>

</div>
```

```
  </div>

  <script type="module">

    import { initializeApp } from
"https://www.gstatic.com/firebasejs/11.6.1/firebase-app.js";

    import { getAuth, signInAnonymously, signInWithCustomToken,
onAuthStateChanged } from
"https://www.gstatic.com/firebasejs/11.6.1/firebase-auth.js";

    import { getFirestore, collection, onSnapshot, doc, setDoc, addDoc } from
"https://www.gstatic.com/firebasejs/11.6.1/firebase-firestore.js";

    const firebaseConfig = (typeof __firebase_config !== 'undefined' &&
__firebase_config)

      ? JSON.parse(__firebase_config)

      : {

          apiKey: "YOUR_API_KEY",

          authDomain: "YOUR_PROJECT_ID.firebaseapp.com",

          projectId: "YOUR_PROJECT_ID",

          storageBucket: "YOUR_PROJECT_ID.appspot.com",

          messagingSenderId: "YOUR_SENDER_ID",

          appId: "YOUR_APP_ID",

        };

    const isPlaceholderFirebaseConfig = !firebaseConfig ||
firebaseConfig.projectId === 'YOUR_PROJECT_ID';

    if (isPlaceholderFirebaseConfig) {

      console.warn('Firebase config is using placeholder values. Replace these
with your actual Firebase project settings.');

    }

   const app = isPlaceholderFirebaseConfig ? null :
initializeApp(firebaseConfig);

    const auth = isPlaceholderFirebaseConfig ? null : getAuth(app);

    const db = isPlaceholderFirebaseConfig ? null : getFirestore(app);

    const appId = typeof __app_id !== 'undefined' ? __app_id : 'employee-
directory-app';
```

```
const localMode = isPlaceholderFirebaseConfig;

const DEPARTMENTS = ["All Departments", "Engineering", "Marketing", "Design", "Operations"];

const FORM_DEPARTMENTS = ["Engineering", "Marketing", "Design", "Operations"];

const SEED_RECORDS = [

  { id: "1", name: "Alice Johnson", dept: "Engineering", joined: "2022-03-15", role: "Frontend Dev" },

  { id: "2", name: "Bob Smith", dept: "Marketing", joined: "2021-11-02", role: "SEO Specialist" },

  { id: "3", name: "Charlie Davis", dept: "Engineering", joined: "2023-01-20", role: "Backend Dev" },

  { id: "4", name: "Diana Prince", dept: "Design", joined: "2022-08-12", role: "UI Designer" },

  { id: "5", name: "Ethan Hunt", dept: "Operations", joined: "2020-05-30", role: "Manager" },

];

let currentUser = null;

let records = [];

let searchTerm = "";

let selectedDept = "All Departments";

let sortConfig = { key: 'name', direction: 'asc' };

let reportFileName = "staff_directory_report.json"

const errorDiv = document.getElementById('errorDiv');

const errorText = errorDiv.querySelector('span.text-sm');

const tableBody = document.getElementById('tableBody');

const tableContainer = document.getElementById('tableContainer');

const searchInput = document.getElementById('searchInput');

const deptSelect = document.getElementById('deptSelect');

const fileNameInput = document.getElementById('fileNameInput');

const totalStaffCount = document.getElementById('totalStaffCount');
```

```
const statsContainer = document.getElementById('statsContainer');
const footerStats = document.getElementById('footerStats');
const exportBtn = document.getElementById('exportBtn');
const connectionStatus = document.getElementById('connectionStatus');
const nameHeaderBtn = document.getElementById('nameHeaderBtn');
const joinedHeaderBtn = document.getElementById('joinedHeaderBtn');
const nameArrow = document.getElementById('nameArrow');
const joinedArrow = document.getElementById('joinedArrow');
const addEmployeeBtn = document.getElementById('addEmployeeBtn');
const modal = document.getElementById('modal');
const modalForm = document.getElementById('modalForm');
const cancelBtn = document.getElementById('cancelBtn');
const nameInput = document.getElementById('nameInput');
const deptInput = document.getElementById('deptInput');
const dateInput = document.getElementById('dateInput');
const roleInput = document.getElementById('roleInput');
const closeModalBtn = document.getElementById('closeModalBtn');

function initAuth() {
  if (localMode) {
    console.warn('Firebase is not configured. Running in local demo mode. Replace with valid config to enable Firestore.');
    if (!records.length) {
      records = [...SEED_RECORDS];
    }
    updateConnectionStatus();
    render();
    return;
  }
```

```
const initAuthAsync = async () => {
  try {
    if (typeof __initial_auth_token !== 'undefined' && __initial_auth_token) {
      await signInWithCustomToken(auth, __initial_auth_token);
    } else {
      await signInAnonymously(auth);
    }
  } catch (err) {
    console.error('Auth failed:', err);
    showError('Database authentication failed. Please refresh.');
  }
};
initAuthAsync();

onAuthStateChanged(auth, (user) => {
  currentUser = user;
  updateConnectionStatus();
  if (user) setupDataListener();
});   }
function updateConnectionStatus() {
 const statusDot = connectionStatus.querySelector('.status-dot');
 const statusText = connectionStatus.querySelector('.status-text');
if (localMode) {
  statusDot.className = 'status-dot w-1.5 h-1.5 rounded-full bg-amber-500';
  statusText.textContent = 'Demo Mode';
  return;
}
if (!currentUser) {
  statusDot.className = 'status-dot w-1.5 h-1.5 rounded-full bg-amber-500';
```

```
      statusText.textContent = 'Connecting';
    } else {
      statusDot.className = 'status-dot w-1.5 h-1.5 rounded-full bg-emerald-
500';
      statusText.textContent = 'Connected';
    }
  }
  function setupDataListener() {
    if (localMode) {
      if (!records.length) {
        records = [...SEED_RECORDS];
      }
      render();
      return;
    }
    const collectionRef = collection(db, 'artifacts', appId, 'public', 'data',
'employees');
    onSnapshot(collectionRef, (snapshot) => {
      if (snapshot.empty) {
        SEED_RECORDS.forEach(async (record) => {
          try {
            const docRef = doc(db, 'artifacts', appId, 'public', 'data', 'employees',
record.id);
            await setDoc(docRef, record);
          } catch (e) {
            console.error('Seeding error:', e);
          }
        });
      } else {
        records = snapshot.docs.map(doc => ({ ...doc.data(), id: doc.id }));
```

```
      render();
    }
  }, (err) => {
    console.error('Firestore error:', err);
    showError('Unable to sync with cloud. Check your connection.');
  });
}
function setupEventListeners() {
  searchInput.addEventListener('input', (e) => {
    searchTerm = e.target.value;
    render();
  });\
  deptSelect.addEventListener('change', (e) => {
    selectedDept = e.target.value;
    render();
  });

  fileNameInput.addEventListener('input', (e) => {
    reportFileName = e.target.value;
  });\
  exportBtn.addEventListener('click', handleExport);
  nameHeaderBtn.addEventListener('click', () => requestSort('name'));
  joinedHeaderBtn.addEventListener('click', () => requestSort('joined'));
  addEmployeeBtn.addEventListener('click', openModal);
  closeModalBtn.addEventListener('click', closeModal);
  cancelBtn.addEventListener('click', closeModal);
  modalForm.addEventListener('submit', handleAddEmployee);
  modal.addEventListener('click', (e) => {
    if (e.target === modal) closeModal();
```

```
  });
}
function populateDepartments() {
  DEPARTMENTS.forEach((dept) => {
    const option = document.createElement('option');
    option.value = dept;
    option.textContent = dept;
    deptSelect.appendChild(option);
  });\
  FORM_DEPARTMENTS.forEach((dept) => {
    const option = document.createElement('option');
    option.value = dept;
    option.textContent = dept;
    deptInput.appendChild(option);
  });
}

function openModal() {
  modal.classList.remove('hidden');
  nameInput.value = '';
  roleInput.value = '';
  deptInput.value = 'Engineering';
  dateInput.value = new Date().toISOString().split('T')[0];
}
function closeModal() {
  modal.classList.add('hidden');
}\
async function handleAddEmployee(event) {
  event.preventDefault();
```

```
    if (localMode) {
      const newEmployee = {
        id: `${Date.now()}`,
        name: nameInput.value.trim(),
        dept: deptInput.value,
        role: roleInput.value.trim(),
        joined: dateInput.value,
      };
      records.push(newEmployee);
      closeModal();
      render();
      return;
    }
    if (!currentUser) return;
    const newEmployee = {
      name: nameInput.value.trim(),
      dept: deptInput.value,
      role: roleInput.value.trim(),
      joined: dateInput.value,
    };
    try {
      const collectionRef = collection(db, 'artifacts', appId, 'public', 'data',
'employees');
      await addDoc(collectionRef, newEmployee);
      closeModal();
    } catch (err) {
      console.error('Error adding employee:', err);
      showError('Failed to add employee. Please try again.');
    }
```

```
  }
  function requestSort(key) {
    sortConfig = {
      key,
      direction: sortConfig.key === key && sortConfig.direction === 'asc' ?
'desc' : 'asc',
    };
    render();
  }
  function getProcessedRecords() {
    let result = [...records];
    if (searchTerm) {
      const term = searchTerm.toLowerCase();
      result = result.filter((r) => r.name.toLowerCase().includes(term) ||
r.role.toLowerCase().includes(term));
    }
    if (selectedDept !== 'All Departments') {
      result = result.filter((r) => r.dept === selectedDept);
    }
    result.sort((a, b) => {
      const valA = (a[sortConfig.key] || '').toString().toLowerCase();
      const valB = (b[sortConfig.key] || '').toString().toLowerCase();
      if (valA < valB) return sortConfig.direction === 'asc' ? -1 : 1;
      if (valA > valB) return sortConfig.direction === 'asc' ? 1 : -1;
      return 0;
    });
    return result;
  }
  function getDeptStats() {
    const counts = {};
```

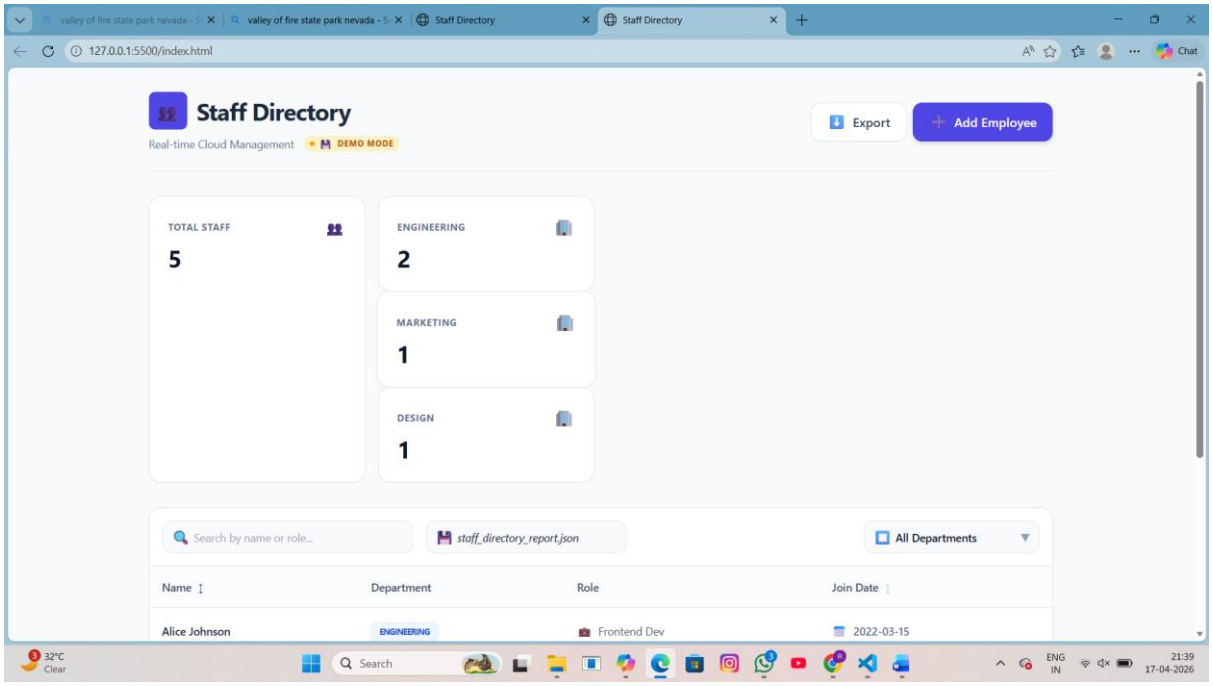
```
    records.forEach((r) => {
      counts[r.dept] = (counts[r.dept] || 0) + 1;
    });
    return Object.entries(counts).map(([name, count]) => ({ name, count }));
  }
  function getDeptBadgeColor(dept) {
    if (dept === 'Engineering') return 'bg-blue-50 text-blue-700 border-blue-100';
    if (dept === 'Marketing') return 'bg-purple-50 text-purple-700 border-purple-
100';
    if (dept === 'Design') return 'bg-pink-50 text-pink-700 border-pink-100';
    return 'bg-orange-50 text-orange-700 border-orange-100';
  }
  function getDeptDotColor(name) {
    if (name === 'Engineering') return 'bg-blue-400';
    if (name === 'Marketing') return 'bg-purple-400';
    if (name === 'Design') return 'bg-pink-400';
    return 'bg-orange-400';
  }
  function render() {
    const processed = getProcessedRecords();
    const stats = getDeptStats();
    nameArrow.className = sortConfig.key === 'name' ? 'text-indigo-600 text-
sm' : 'text-slate-300 group-hover:text-slate-400 text-sm';
    joinedArrow.className = sortConfig.key === 'joined' ? 'text-indigo-600
text-sm' : 'text-slate-300 group-hover:text-slate-400 text-sm';
    tableBody.innerHTML = '';
    if (processed.length === 0) {
      tableBody.innerHTML = '<tr><td colspan="4" class="p-20 text-
center"><div class="flex flex-col items-center gap-3 text-slate-400"><div
class="bg-slate-100 p-4 rounded-full text-2xl">🔍</div><p class="font-medium
text-sm">No results match your current filters.</p><button
```

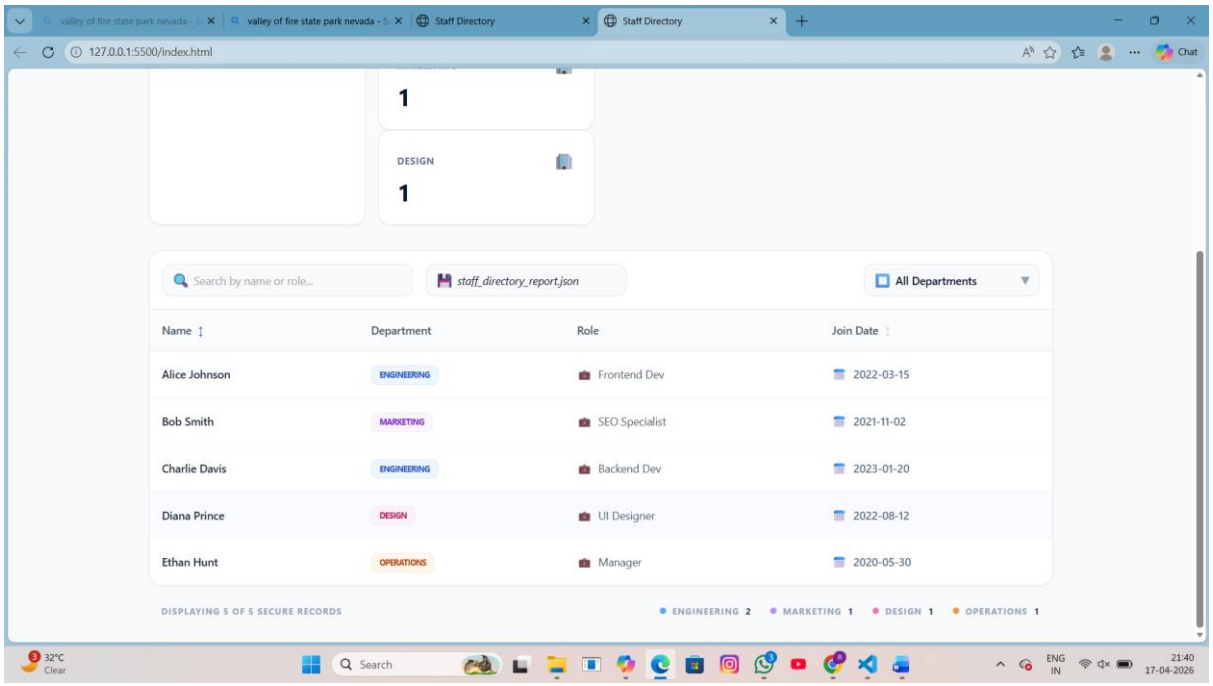
```
id="clearFiltersBtn2" class="text-indigo-600 text-sm font-bold
hover:underline">Clear all filters</button></div></td></tr>';

      document.getElementById('clearFiltersBtn2').addEventListener('click', ()
=> {

        searchTerm = '';

        selectedDept = 'All Departments';

        searchInput.value = '';

        deptSelect.value = 'All Departments';

        render();

      });

    } else {

      processed.forEach((record) => {

        const row = document.createElement('tr');

        row.className = 'hover:bg-indigo-50/30 transition-colors group border-b
border-slate-100';

        row.innerHTML = `

          <td class="p-4 font-semibold text-slate-800 text-
sm">${escapeHtml(record.name)}</td>

          <td class="p-4"><span class="inline-flex items-center px-2.5 py-1
rounded-lg text-[10px] font-bold uppercase tracking-tight border
${getDeptBadgeColor(record.dept)}">${escapeHtml(record.dept)}</span></td>

          <td class="p-4 text-slate-600"><div class="flex items-center gap-2 text-
sm"><span class="text-slate-
400">💼</span>${escapeHtml(record.role)}</div></td>

          <td class="p-4 text-slate-500"><div class="flex items-center gap-2 text-
sm font-medium"><span class="text-slate-
400">📅</span>${escapeHtml(record.joined)}</div></td>

        `;

        tableBody.appendChild(row);

      });

    }

    totalStaffCount.textContent = records.length;
```

```
    statsContainer.innerHTML = '';
    stats.slice(0, 3).forEach((stat) => {
      const div = document.createElement('div');
      div.className = 'bg-white p-6 rounded-2xl border border-slate-200
shadow-sm transition-transform hover:scale-[1.01]';
      div.innerHTML = `<div class="flex items-center justify-between mb-
2"><span class="text-slate-500 text-xs font-bold uppercase tracking-
wider">${escapeHtml(stat.name)}</span><span class="text-emerald-500 text-
xl">🏢</span></div><div class="text-3xl font-bold">${stat.count}</div>`;
      statsContainer.appendChild(div);
    });
    footerStats.innerHTML = '';
    stats.forEach((s) => {
      const div = document.createElement('div');
      div.className = 'flex items-center gap-2';
      div.innerHTML = `<span class="w-2 h-2 rounded-full
${getDeptDotColor(s.name)}"></span><span>${escapeHtml(s.name)} <span
class="text-slate-700 ml-1">${s.count}</span></span>`;
      footerStats.appendChild(div);
    });
    document.getElementById('processingInfo').textContent = `Displaying
${processed.length} of ${records.length} Secure Records`;
  }
  function showError(message) {
    errorText.textContent = message;
    errorDiv.classList.remove('hidden');
  }
  function handleExport() {
    const processed = getProcessedRecords();
    const dataStr = JSON.stringify(processed, null, 2);
    const dataUri = 'data:application/json;charset=utf-8,' +
encodeURIComponent(dataStr);
```

```
    const linkElement = document.createElement('a');
    linkElement.setAttribute('href', dataUri);
    linkElement.setAttribute('download', reportFileName || 'export.json');
    linkElement.click();
  }
  function escapeHtml(text) {
    const div = document.createElement('div');
    div.textContent = text;
    return div.innerHTML;
  }

  window.addEventListener('DOMContentLoaded', () => {
    setupEventListeners();
    populateDepartments();
    initAuth();
    render();
  });
 </script>
</body>
</html>
```

Output:


Staff Directory
Real-time Cloud Management
DEMO MODE
Export
Add Employee
TOTAL STAFF
5
ENGINEERING
2
MARKETING
1
DESIGN
1
Search by name or role...
staff_directory_report.json
All Departments
Name
Department
Role
Join Date
Alice Johnson
ENGINEERING
Frontend Dev
2022-03-15


1
DESIGN
1
Search by name or role...
staff_directory_report.json
All Departments
Name
Department
Role
Join Date
Alice Johnson
ENGINEERING
Frontend Dev
2022-03-15
Bob Smith
MARKETING
SEO Specialist
2021-11-02
Charlie Davis
ENGINEERING
Backend Dev
2023-01-20
Diana Prince
DESIGN
UI Designer
2022-08-12
Ethan Hunt
OPERATIONS
Manager
2020-05-30
DISPLAYING 5 OF 5 SECURE RECORDS
ENGINEERING 2
MARKETING 1
DESIGN 1
OPERATIONS 1

Result :

A functional data retrieval and sorting dashboard was successfully developed. The system displays student/employee records in a structured tabular format, making the data easy to read and manage. It provides efficient sorting options based on name and date, allowing users to quickly organize information as needed.